2026

Introduction to Computational Biology

Instructor: Prof. Rachel Kolodny

Office Hours: Coordinate by email: trachel@gmail.com

Course Type: Lecture

Course Level: B.Sc.+M.Sc.

Course Overview:

The course presents methods and algorithms for modeling and solving problems in Biology, with an emphasis on molecular biology. The course includes algorithms for sequence and structure alignment, application of methods from computational learning theory, and specifically methods for addressing the problem of protein structure prediction. All background needed in Biology is covered within the class.

Topics:

1. Introduction to Biology:
DNA, RNA, Proteins, Basic concepts of protein structure

2. Algorithms for Sequence Alignment:
Pairwise alignments: Needlman Wunch, Smith-Waterman, Linear space algorithms
Constructing substitution matrices
FASTA and BLAST/PSI-BLAST heuristics, and the estimation of the significance of pairwise alignments
Hirschberg's algorithm
Multiple sequence alignment.

3. HMM aligners (and HMMs in general)
Forward-Backward algorithm
Viterbi's algorithm
Baum-Weltch algorithm

4. Algorithms for manipulating objects in three dimensions:
Description of rotation and translation in Robotics (including quaternions)

Finding the optimal rotation and translation for superimposing two sets of points in 3D
Calculating 3D structures from the internal distance matrices (SVD)

5. Algorithms for structure alignment

6. Methods from computational learning
HMMs
Neural nets for secondary structure prediction (just an intro)

7. Predicting protein structure using AI

8. Introduction to Protein Language Models

9. Other topics (if time permits)

Requirements: Exam 90%, all assignments (3) 10%